



## Technical Reproducibility Validation Report

### HipSTR

Version v0.7  
Autosomal STR

|                   |                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Verification date | 2026-09-19                                                                                                                            |
| Repository commit | <a href="https://github.com/STRhub/hipstr/commit/b2033bfb5cf55496b776463bdf2993fa763a4be">b2033bfb5cf55496b776463bdf2993fa763a4be</a> |
| Attestation level | <b>Runs + Plausible output</b>                                                                                                        |
| Permalink         | <a href="https://strhub.app/verified/hipstr">https://strhub.app/verified/hipstr</a>                                                   |
| STRhub            | <a href="https://strhub.app">https://strhub.app</a>                                                                                   |

#### SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

#### WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

#### NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance



## 1. Executive Summary

|                   |                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------|
| PURPOSE           | Verify that the tool installs, runs end-to-end, and produces structurally valid output. |
| RESULT            | 5/5 gates passed. Runs + Plausible output                                               |
| ATTESTATION LEVEL | Runs + Plausible output                                                                 |
| SCOPE             | Installation, execution, and output structure verification.                             |
| NOT EVALUATED     | Genotype accuracy · Concordance · Forensic validity · Regulatory compliance             |

## 2. Reproducibility Metadata

|                 |                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tool            | HipSTR                                                                                                                                                |
| Version         | v0.7                                                                                                                                                  |
| Repository      | <a href="https://github.com/tfwillems/HipSTR">https://github.com/tfwillems/HipSTR</a>                                                                 |
| Commit (pinned) | <a href="https://github.com/tfwillems/HipSTR/commit/b2033bfb5cf55496b776463bdf2993fa763a4be">b2033bfb5cf55496b776463bdf2993fa763a4be</a>              |
| Container OS    | ubuntu-22.04                                                                                                                                          |
| Marker panel    | Autosomal STR                                                                                                                                         |
| Recipe          | a recipe STRhub wrote by hand, not read off the repository: a note under the documented result, never the badge                                       |
| Verified on     | 2026-09-19 18:22:28 UTC                                                                                                                               |
| CI run          | <a href="https://github.com/Tfronta/strhub-verified/actions/runs/35460767865">https://github.com/Tfronta/strhub-verified/actions/runs/35460767865</a> |

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

## 3. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · HipSTR v0.7 · commit b2033bf
$ HipSTR \
--bams /data/in/input.bam \
--fasta /data/ref/hg38.fa \
--regions /data/in/regions.bed \
--str-vcf /data/out/result.vcf.gz \
--min-reads 5 \
--use-unpaired \
--read-qual-trim ! \
--def-stutter-model \
--max-str-len 250
```

## 4. Verification Gates

|                  |                                                           |      |
|------------------|-----------------------------------------------------------|------|
| Available        | The pinned public source exists                           | PASS |
| Installs         | The environment builds from source                        | PASS |
| Runs             | Executes end-to-end without crashing                      | PASS |
| Expected IO      | Produces a non-empty file in the declared format          | PASS |
| Plausible Output | Output structure matches expected genotype-bearing format | PASS |
| Execution log    | <a href="#">hipstr.log-external.txt</a>                   |      |

## 5. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

## 6. Output Content Evidence

|                          |               |
|--------------------------|---------------|
| Output file              | result.vcf.gz |
| Format                   | VCF           |
| Sequence records         | 23            |
| Distinct STR loci        | 23            |
| Total reads              | 8,230         |
| Max depth (single locus) | 523           |

## 7. Verification Data

The repository ships example data (3 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

|         |                                                                                                                                               |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset | Illumina BAM (hg38), NA12878 (autosomal, female)                                                                                              |
| Source  | <a href="https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat...">https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat...</a> |
| License | Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.                                                     |

|             |                                                                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ref. genome | GRCh38 / hg38                                                                                                                                                                                                 |
| Loci tested | CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D5S818 · D6S1043 · D7S820 · D8S1179 · FGA · PentaD · PentaE · SE33 · TH01 · TPOX · vWA |
| Regions BED | Provided by STRhub                                                                                                                                                                                            |

## 8. Verification Matrix

|             |                             |                                                  |
|-------------|-----------------------------|--------------------------------------------------|
| <b>PASS</b> | Reference dataset           | Illumina BAM (hg38), NA12878 (autosomal, female) |
|             | README check (5/5)          | Advisory: all items present                      |
|             | Install / environment setup | Present                                          |
|             | Run command                 | Present                                          |
|             | Expected input format       | Present                                          |
|             | Produced output format      | Present                                          |
|             | Dependencies / versions     | Present                                          |

## 9. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

- A container environment and a command written by STRhub, not taken from the repository's instructions; what they do differently is listed under the recipe.

## 10. What STRhub's recipe does that the repository's instructions do not

This run used a recipe STRhub wrote. Each item below is a departure from the README that a first-time user would have to discover for themselves. The badge rests on the run of the repository's own instructions, not on this one.

- Runs HipSTR with flags tuned to STRhub's reference slice (--min-reads 5 --use-unpaired --read-quality-trim ! --def-stutter-model --max-str-len 250). — instead of: The README's command, which has none of them. On a BAM slice around the panel loci most mates fall outside the slice and read quality trims most reads; without these flags nearly every locus is skipped.
- Supplies the regions BED from STRhub's library. — instead of: A regions file the README leaves to the user.

## 11. Limitations

- Single reference dataset per input type
- Single containerized environment (Docker / ubuntu-22.04)
- No truth-set comparison or ground-truth genotypes
- No accuracy or concordance assessment
- No forensic validation of results
- Short-read limitations apply (very long STR alleles may not span reads)

## 12. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

## 13. Conclusion

### ◆ Runs end-to-end

HipSTR v0.7 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

### ◆ Produces plausible output

The tool generated a structurally valid VCF output file with genotype calls across 23 target forensic STR loci, with a maximum read depth of 523 at D18S51.

### ◆ Verdict: Runs

The tool installed and its run produced its documented output.

### ◆ Repository test data not used

The repository ships example data (3 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

### ◆ Reproducibility statement

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.



| LOCUS    | READ DEPTH |                      | LOCUS    | READ DEPTH |                |
|----------|------------|----------------------|----------|------------|----------------|
| D18S51   | 523        | ████████████████████ | D12S391  | 361        | ██████████████ |
| FGA      | 475        | ████████████████████ | D1S1656  | 360        | ██████████████ |
| vWA      | 465        | ████████████████████ | D8S1179  | 349        | ██████████████ |
| PentaD   | 444        | ████████████████████ | D13S317  | 321        | ██████████████ |
| D2S441   | 433        | ████████████████████ | D10S1248 | 313        | ██████████████ |
| D3S1358  | 423        | ████████████████████ | D19S433  | 300        | ██████████████ |
| D22S1045 | 416        | ████████████████████ | CSF1PO   | 280        | ██████████████ |
| D7S820   | 412        | ████████████████████ | TH01     | 278        | ██████████████ |
| PentaE   | 408        | ████████████████████ | D2S1338  | 265        | ██████████████ |
| D5S818   | 400        | ████████████████████ | TPOX     | 264        | ██████████████ |
| D16S539  | 374        | ██████████████       | SE33     | 0          |                |
| D6S1043  | 366        | ██████████████       |          |            |                |